

N+- Network Fundamentals and building network

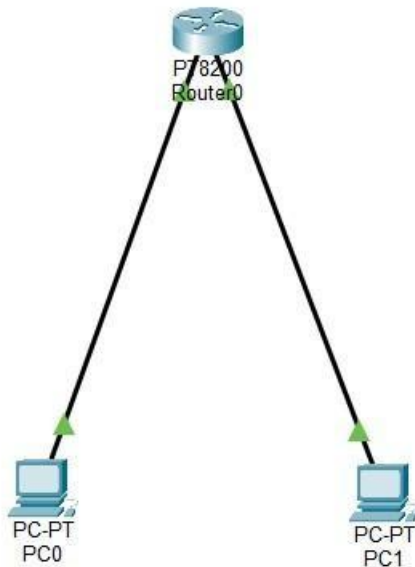
[Q] Review and Practice Practical (1 hour): Create a physical network topology in Cisco Packet Tracer. Task: Document the steps for configuring a small network.

[Q-1] Open Cisco Packet Tracer and create a new project file named 'HomeWiFiNetwork.pkt' with three PCs and one wireless router connected appropriately.

[ANS]

○ Steps

1. Open Cisco Packet Tracer.
2. Create a new project.
3. Drag and drop one **Wireless Router** into the workspace.
4. Drag and drop **three PCs** into the workspace.
5. Connect each PC to the wireless router using Copper Straight-Through cables.
6. Save the project as **HomeWiFiNetwork.pkt**.



[Q-2] Assign static IP addresses to each PC in your Packet Tracer network so they are on the same subnet (e.g., 192.168.10.x), and configure the default gateway to the router's IP.

[ANS]

○ Router Configuration

Device	IP Address
Wireless Router	192.168.10.1

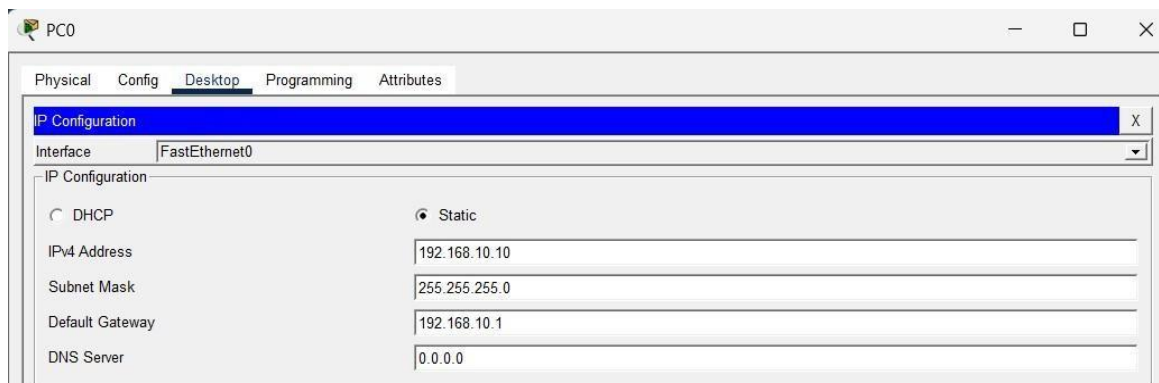
○ PC IP Configuration

Device	IP Address	Subnet Mask	Default Gateway
PC1	192.168.10.10	255.255.255.0	192.168.10.1

PC2	192.168.10.20	255.255.255.0	192.168.10.1
PC3	192.168.10.30	255.255.255.0	192.168.10.1

○ Steps

1. Click on PC1.
2. Go to **Desktop → IP Configuration**.
3. Enter:
 - IP Address: 192.168.10.10
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.10.1
4. Repeat for PC2 and PC3 using their assigned IP addresses.



**[Q-3] Test connectivity by using the Packet Tracer 'ping' tool to verify that all PCs can communicate with each other and the router.

Hint: Use the Command Prompt tool on each PC and type 'ping [other PC's IP]'. **

[ANS]

○ Commands Used

- **From PC1** ping
192.168.10.20 ping
192.168.10.30 ping

192.168.10.1

- **From PC2** ping

192.168.10.10 ping

192.168.10.30 ping

192.168.10.1

- **From PC3** ping

192.168.10.10 ping

192.168.10.20 ping

192.168.10.1

○ Sample Result

Reply from 192.168.10.20: bytes=32 time<1ms TTL=128

Reply from 192.168.10.30: bytes=32 time<1ms TTL=128

Reply from 192.168.10.1: bytes=32 time<1ms TTL=128

○ Conclusion

All PCs successfully communicated with each other and with the router, confirming proper network configuration.

```
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

[Q-4] Document each step you took to build and configure your Packet Tracer network, including screenshots of your topology and the IP configuration windows for each PC, in a Word or PDF file.

[ANS]

- **Step 1: Create the Network •**

Open Cisco Packet Tracer.

- Add one Wireless Router.
- Add three PCs.
- Connect all PCs to the router.

○ Step 2: Configure Router

- Open the router configuration.
- Assign IP address 192.168.10.1.

○ Step 3: Configure PC1

- Desktop → IP Configuration.
- IP Address: 192.168.10.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.10.1

○ Step 4: Configure PC2

- Desktop → IP Configuration.
- IP Address: 192.168.10.20
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.10.1

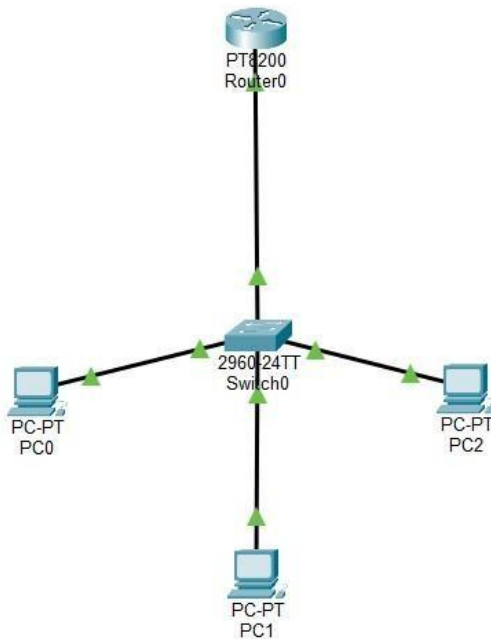
○ Step 5: Configure PC3

- Desktop → IP Configuration.
- IP Address: 192.168.10.30
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.10.1

○ Step 6: Verify Connectivity

- Open Command Prompt on each PC.

- Use the ping command to test communication.
- Confirm successful replies from all devices.



PC0

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.10.10

Subnet Mask 255.255.255.0

Default Gateway 192.168.10.1

DNS Server 0.0.0.0

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.10.20

Subnet Mask 255.255.255.0

Default Gateway 192.168.10.1

DNS Server 0.0.0.0

PC2

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.10.30

Subnet Mask 255.255.255.0

Default Gateway 192.168.10.1

DNS Server 0.0.0.0